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SUPERSEDING
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20 June 1963

MILITARY SPECIFICATION

CAPACITORS, FIXED, MICA DIELECTRIC, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

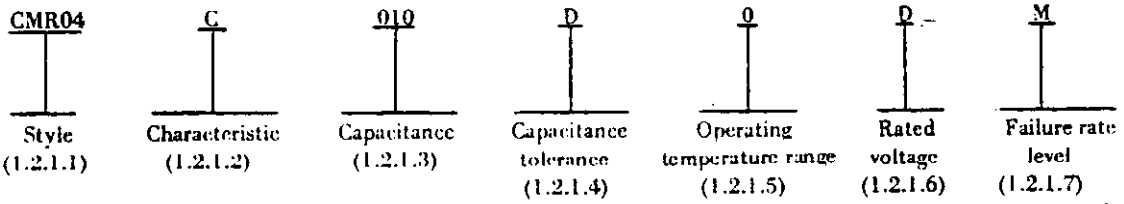
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for established reliability, mica dielectric, fixed capacitors (see 6.1). Capacitors covered by this specification have a failure rate level ranging from 2.0 percent per 1,000 hours to 0.0001 percent per 1,000 hours (see 1.2.1.7). The failure rate level is established at a 90-percent confidence level. The failure rate level, identified by the applicable symbol specified in table V, is referred to operational life at full rated voltage (see 1.2.1.6) at 125° or 150° C, as applicable. An acceleration factor of 25:1 has been used to relate life test data obtained at 150 percent of rated voltage at rated temperature to rated voltage at rated temperature.

1.2 Classification.

1.2.1 *Type designation.* The type designation shall be in the following form, and as specified (see 3.1 and 6.2):



1.2.1.1 *Style.* The style is identified by the three-letter symbol "CMR" followed by a two-digit number; the letters identify mica dielectric, fixed capacitors, established reliability, and the number identifies the shape and dimensions of the capacitor.

1.2.1.2 *Characteristic.* The characteristic is identified by a single letter which indicates the relative stability of the capacitor with temperature change, in accordance with table I.

TABLE I. *Characteristic.*

Symbol	Temperature coefficient	Capacitance drift
	Parts/million/°C	
C	-200 to +200	±(0.5 percent +0.1 pF)
E	-20 to +100	±(0.1 percent +0.1 pF)
F	0 to +70	±(0.05 percent +0.1 pF)

1.2.1.3 *Capacitance.* The nominal capacitance value expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

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1.2.1.4 *Capacitance tolerance.* The capacitance tolerance in percent or pF, as applicable, is identified by a single letter in accordance with table II.

TABLE II. *Capacitance tolerance.*

Symbol	Capacitance tolerance (±)
D	0.5 pF
F	1 percent
G	2 percent
J	5 percent

1.2.1.5 *Operating temperature range.* The operating temperature range is identified by a single letter in accordance with table III.

TABLE III. *Operating temperature range.*

Symbol	Operating temperature range °C
O	-55 to +125
P	-55 to +150

1.2.1.6 *Rated voltage.* The rated voltage is identified by a single letter in accordance with table IV.

TABLE IV. *Rated voltage.*

Symbol	DC rated voltage
A	100
C	300
D	500

1.2.1.7 *Failure rate level.* The failure rate level in percent per 1,000 hours is identified by a single letter in accordance with table V, and is based on rated voltage at 125° or 150° C, as applicable.

TABLE V. *Failure rate level.*

Symbol	Failure rate level (percent per 1,000 hours)
L	2.0
M	1.0
P	0.1
R	0.01
S	0.001
T	0.0001

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

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SPECIFICATIONS

MILITARY

MIL-C-39001/5 Capacitors, Fixed, Mica Dielectric, Established Reliability; Styles CMR04, CMR05, CMR06, CMR07 and CMR08.
MIL-C-39028 Capacitors, Packaging of.

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 Test Methods for Electronic and Electrical Component Parts.
MIL-STD-690 Failure Rate Sampling Plans and Procedures.
MIL-STD-790 Reliability Assurance Program for Electronic Parts Specifications.
MIL-STD-810 Environmental Test Methods.
MIL-STD-1285 Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Specification sheet. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet the latter shall govern (see 6.2).

3.2 Qualification. Capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Reliability. Reliability of capacitors furnished under this specification shall be established and maintained in accordance with the procedures and requirements specified in MIL-STD-790 and MIL-STD-690 with details specified in 4.1.1, 4.4.4 and 4.5.

3.4 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 *Insulating, impregnating, and sealing compounds.* Insulating, impregnating, and sealing compounds, including varnishes, waxes, and the like, shall be suitable for each particular application. Compounds shall preserve the electrical characteristics of the insulation to which applied. The compound, either in the state of its original application or as a result of cracking or aging, shall have no adverse effect on the performance of the capacitor.

3.4.2 *Terminal leads.* Leads shall be made of a solid conductor of the length and diameter specified (see 3.1), and shall be coated with solder having a tin content of 40 to 75 percent and shall meet the solderability requirements of 3.12.

3.5 Design and construction. Capacitors shall be of the design, construction, and physical dimensions specified (see 3.1).

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3.5.1 *Body enclosure.* Capacitors shall be effectively sealed against the entry of moisture, and the elements shall be mounted so as to prevent injurious movement in the capacitor body enclosures. The capacitor elements shall be completely enclosed by the body enclosure.

3.5.2 *Connections.* Internal electrical connections shall be secured by pressure and not depend upon the case material for maintenance of contact over the operating temperature range. Connections between the terminal clip and the active element may additionally be secured by soldering or welding. The connection between the lead wire and the terminal clip shall be soldered or welded.

3.6 **Dielectric withstanding voltage.** Capacitors shall withstand the direct-current (dc) potential specified in 4.7.2 without damage, arcing, or breakdown.

3.7 **Barometric pressure.** Capacitors shall withstand the dc potential specified in 4.7.3 without damage, arcing, or breakdown.

3.8 **Insulation resistance.**

3.8.1 *At room ambient temperature.* When measured as specified in 4.7.4.1, the insulation resistance shall be not less than the applicable value specified on figure 1 or table VI.

3.8.2 *At high ambient temperature.* When measured as specified in 4.7.4.2, the insulation resistance shall be not less than the applicable value specified on figure 1 or table VI.

TABLE VI. *Insulation resistance.*

Capacitance rating	Minimum insulation resistance
At 25° C:	
0 to 10,000 picofarads	100,000 megohms
10,000 picofarads and greater	1,000 megohm-microfarads 1/
At 125° C:	
0 to 3,300 picofarads	10,000 megohms
3,300 picofarads and greater	33 megohm-microfarads
At 150° C:	
0 to 1,500 picofarads	5,000 megohms
1,500 picofarads and greater	7.5 megohm-microfarads

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

3.9 **Capacitance.** When measured as specified in 4.7.5, the capacitance shall be within the tolerance shown in the type designation (see 3.1).

3.10 **Dissipation factor.** When measured as specified in 4.7.6, the dissipation factor shall not exceed the applicable value shown on figure 2.

3.11 **High voltage stabilization.** When capacitors are tested as specified in 4.7.7, there shall be no evidence of damage, arcing, or breakdown; and the insulation resistance shall be not less then the applicable value specified on figure 1 or table VI.

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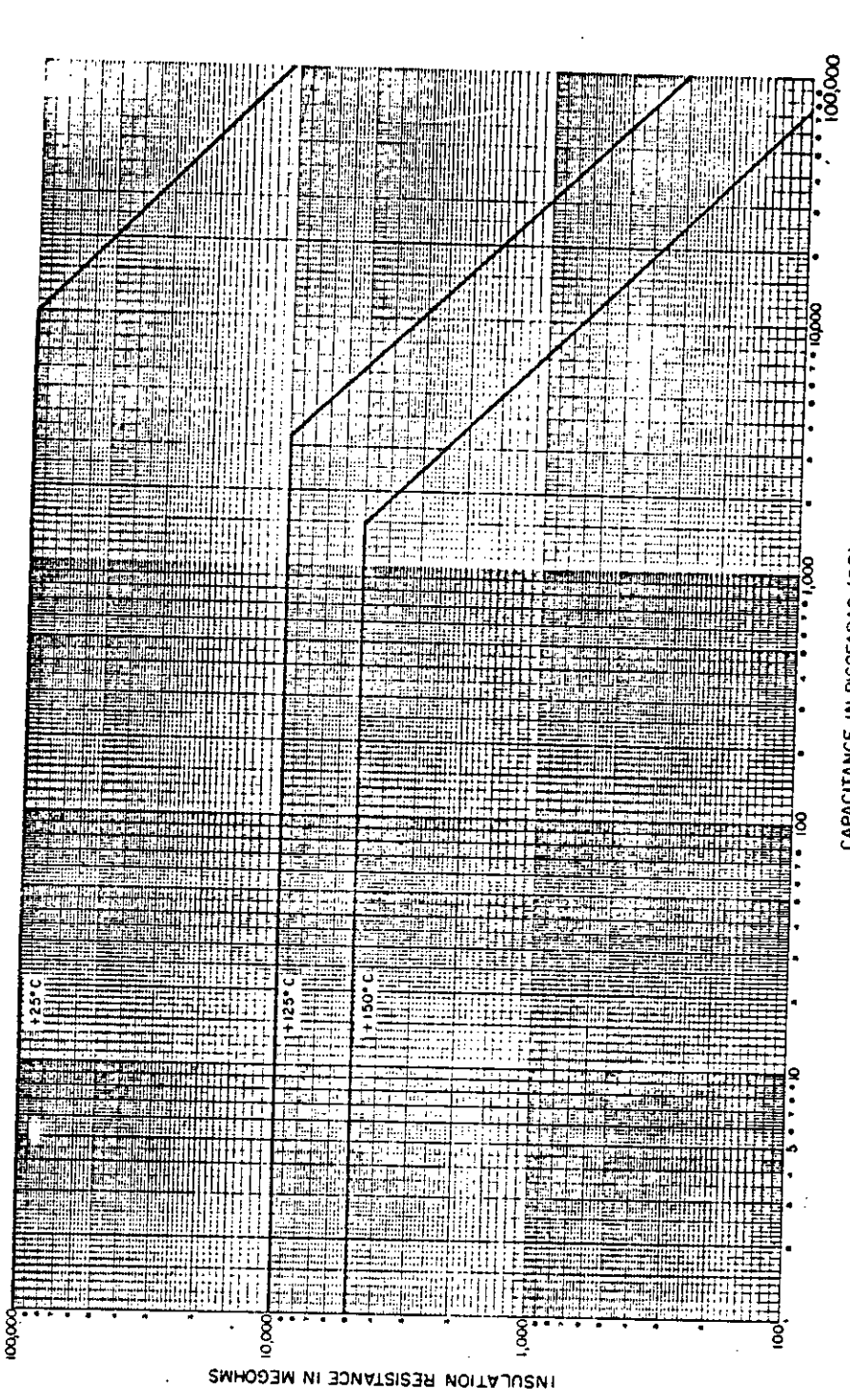


FIGURE 1. Insulation resistance vs capacitance at test temperatures.

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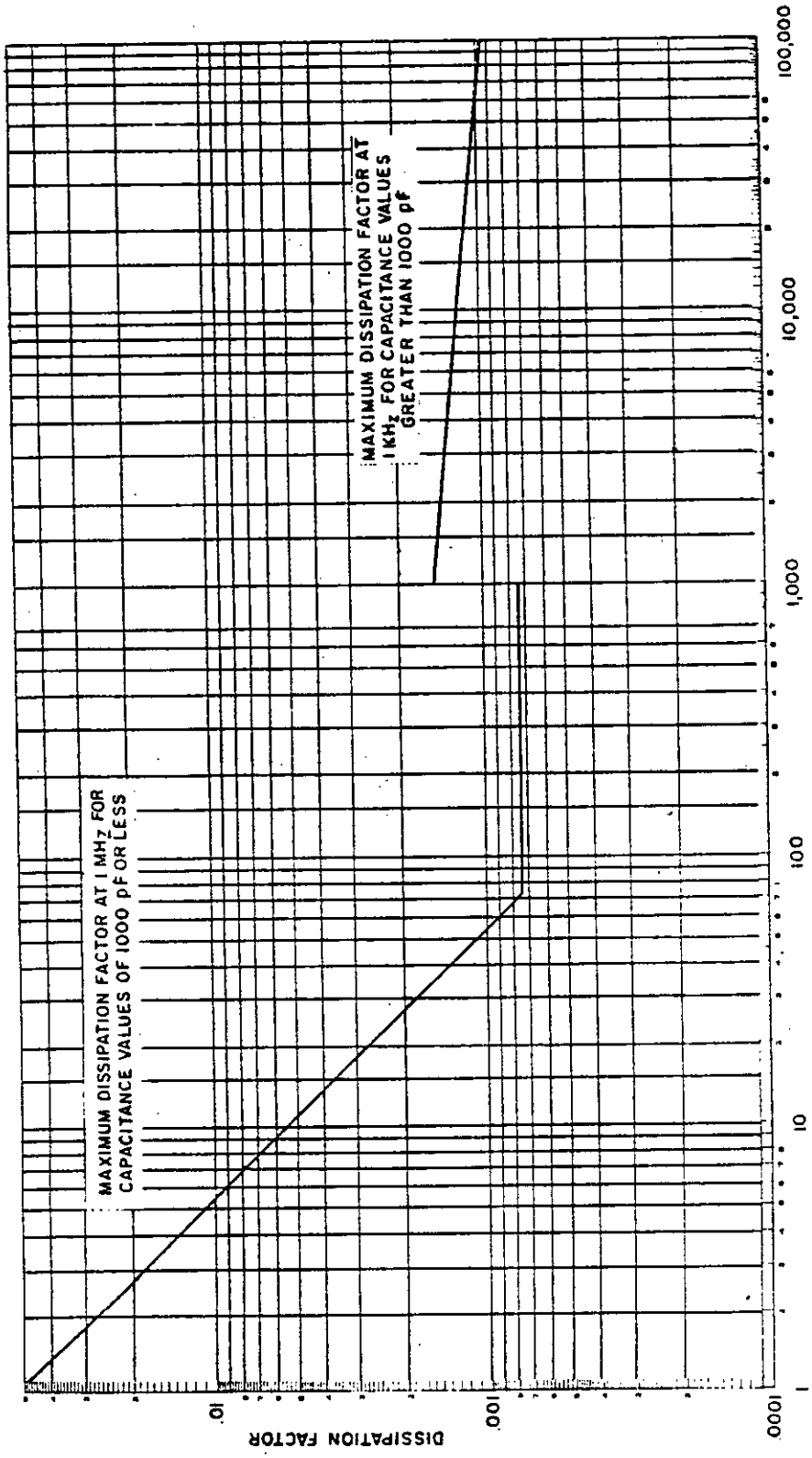


FIGURE 2. Capacitance vs dissipation factor.

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3.12 Solderability. When capacitors are tested as specified in 4.7.8, the dipped surface of the lead shall be at least 95-percent covered with continuous new solder coating. The remaining 5 percent of the lead surface shall show only small pinholes or voids. These shall not be concentrated in one area. No individual view of the dipped surface shall show less than 95 percent coverage. Bare base metal and areas where the solder dip failed to cover the original coating are indications of poor solderability, and shall be cause for failure. In case of dispute, the percent of coverage with pinholes or voids shall be determined by actual measurement of these areas, as compared to the total area.

3.13 Vibration, high frequency. When capacitors are tested as specified in 4.7.9, there shall be no intermittent contacts of 0.5 milliseconds(ms) or greater duration, or momentary arcing, or other indication of breakdown, nor shall there be any open- or short-circuiting or evidence of mechanical damage.

3.14 Temperature coefficient and capacitance drift. When measured as specified in 4.7.10, the temperature coefficient and capacitance drift shall be within the limits specified in table I for the characteristic listed (see 3.1).

3.15 Temperature cycling and immersion. When tested as specified in 4.7.11, capacitors shall meet the following requirements:

Dielectric withstanding voltage	As specified in 3.6.
Insulation resistance	Shall be not less than 30 percent of initial requirement (see 3.8).
Capacitance	The capacitance change shall not exceed 1 percent or 1 pF, whichever is greater, from the reference measurement (see 4.3.2.1).
Dissipation factor	Shall not exceed 150 percent of the initial requirement specified in 3.10.
Visual examination	No visible evidence of deterioration, permanent damage to leads or case, or corrosion on leads.

3.16 Shock, specified pulse. When capacitors are tested as specified in 4.7.12, there shall be no intermittent contacts of 0.5 milliseconds (ms) or greater duration, or momentary arcing, or other indication of breakdown, nor shall there be any open- or short-circuiting or evidence of mechanical damage.

3.17 Terminal strength. When capacitors are tested as specified in 4.7.13, the terminals shall not loosen or rupture and no other damage shall result. Chipping of the resinous coating on the leads, in the "R" dimension area only (see 3.1), shall not be considered as a failure.

3.18 Moisture resistance. When tested as specified in 4.7.14, capacitors shall meet the following requirements:

Dielectric withstanding voltage	As specified in 3.6.
Insulation resistance	Shall be not less than 25 percent of initial requirement (see 3.8).
Capacitance	The capacitance change shall not exceed 1 percent or 1 pF, whichever is greater, from the reference measurement (see 4.3.2.1).
Dissipation factor	Shall not exceed 150 percent of the initial requirement specified in 3.10.
Visual examination	No visible evidence of deterioration, permanent damage to the leads or case, nor corrosion on the leads.

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3.19 Life. When tested as specified in 4.7.15, capacitors shall meet the following requirements:

Dielectric withstanding voltage	As specified in 3.6.
Insulation resistance	Shall not be less than the applicable value specified on figure 1 or table VI.
Capacitance	The capacitance change shall not exceed ±1 percent or ±1 pF, whichever is greater, of the initial measurement.
Dissipation factor	Shall not be greater than 150 percent of initial requirement (see 3.10).
Visual examination	No visible evidence of deterioration, permanent damage to leads or case, arcing, or breakdown.

3.20 Fungus. The manufacturer shall certify that all external materials are fungus resistant or shall perform the test specified in 4.7.16. When capacitors are tested as specified in 4.7.16, examination shall disclose no evidence of fungus growth on the external surface of the capacitor.

3.21 Resistance to solvents. When capacitors are tested as specified in 4.7.17, marking shall remain legible and shall not smear or rub off. In addition, there shall be no visible indication of damage or deterioration to the capacitor body.

3.22 Resistance to soldering heat. When tested as specified in 4.7.18, capacitors shall meet the following requirements:

Insulation resistance	Shall not exceed initial requirement (see 3.8).
Capacitance	The capacitance change shall not exceed ±5 percent or ±5 pF, whichever is greater, from the initial measurement
Dissipation factor	Shall not exceed initial requirement.

3.23 Marking. Marking of capacitors shall conform to method I of MIL-STD-1285, and unless otherwise specified (see 3.1), shall include the type designation, "JAN" brand, trademark, source code, date code (inspection lot of 4.6.1.1), rated voltage, capacitance, capacitance tolerance, and highest rated temperature. Capacitors shall be marked as specified (see 3.1).

3.23.1 *JAN and J marking.* The United States Government has adopted, and is exercising legitimate control over, the certification marks "JAN" and "J" respectively, to indicate electrical equipment, namely, resistors, capacitors, electron tubes and the like, procured by, or manufactured for use by, or for the Government in accordance with standard Government specifications. Accordingly, capacitors procured to, and meeting all of, the requirements specified herein and in applicable specification sheets shall bear the certification mark "JAN," except that capacitors too small to bear the certification mark "JAN" shall bear the letter "J." Capacitors furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein and in applicable specification sheet shall not bear "JAN" or "J". In the event a capacitor sample fails to meet the requirements of this specification and the applicable specification sheet, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all capacitors represented by the sample. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark "JAN."

3.24 Supplying to higher FR levels. A manufacturer may supply to all higher failure rate levels than that to which he is qualified. Parts qualified and marked to lower failure rate levels are substitutable for higher failure level parts, and shall not be remarked unless specified in the contract or purchase order (see 6.2).

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3.25 **Workmanship.** Capacitors shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

3.25.1 *Soldering.* Where soldering is employed, in the construction of capacitors, only fluxes with low corrosive properties shall be used, unless it can be shown that corrosive elements have been satisfactorily removed after soldering. All excess flux and solder shall be removed. Solder shall not be used primarily for obtaining mechanical strength. Electrical connections shall be electrically continuous after soldering and, where possible, mechanically secure before soldering.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the manufacturer is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Reliability assurance program.* A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 **Classification of inspections.** The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) Verification of qualification (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.3 **Inspection conditions and methods.**

4.3.1 *Conditions.* Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 *Methods.*

4.3.2.1 *Reference measurements.* When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at 25° ±5° C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.3 *Power supply.* The power supply used for life testing shall have a regulation of ±2 percent or less of the applicable applied test voltage. The power supply used for dc leakage current measurements shall be stabilized to at least ±100 parts per million. No voltage fluctuations shall occur during measurements that would produce a variation in the current measurement.

4.4 **Qualification inspection.** 1/ Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3), on sample units produced with equipment and procedures normally used in production.

1/ The decision as to whether or not the product will be included on the QPL will be made after successful completion of the 2,000-hour life test.

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4.4.1 *Sample size.* The number of sample units comprising a sample of capacitors to be submitted for qualification inspection shall be as specified in table VII, or in the appendix to this specification. The sample shall be taken from a production run and shall be produced with equipment and procedures normally used in production. Each capacitor style shall be qualified separately.

4.4.2 *Inspection routine.* Sample units shall be subjected to the qualification inspection specified in table VII, in the order shown. All sample units shall be subjected to the group I and II tests. These sample units shall then be divided as shown in table VII for groups III to VI, inclusive, and subjected to the tests for their particular group. The units subjected to life test at rated conditions shall be continued on test for 48,000 additional hours.

4.4.3 *Failures.* Failures in excess of those allowed in table VII shall be cause for refusal to grant qualification approval.

4.4.4 *Failure rate (FR) qualification and lot conformance FR inspection.* FR qualification and lot conformance FR inspection, shall be in accordance with the general and detailed requirements of MIL-STD-690 and the following details:

(a) Procedure I - Qualification at the initial FR level. Level "L" (2.0 percent), of FRSP-90 shall apply. Upon successful completion of the 2,000-hour qualification test (group V, table VII), those sample units which have been subjected to rated conditions shall be continued on test for an additional period as specified in 4.7.15.3.

(b) Procedure II - Extension of qualification to lower FR levels. To extend qualification to the "M" (1.0 percent), "P" (0.1 percent), "R" (0.01 percent), "S" (0.001 percent), and "T" (0.0001 percent) FR levels, data from two or more voltages within a style and of similar construction may be combined. For FR levels "R", "S" and "T", two or more styles of similar construction (see 4.6.1.1) may be combined.

(c) Procedure III - Maintenance of FR level qualification. Maintenance period A of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.5(f)).

(d) Procedure IV - Lot conformance FR inspection. Lot conformance FR procedures shall be in accordance with 4.6.1.3.1 or 4.6.1.3.2, as applicable.

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TABLE VII. *Qualification inspection.*

Examination or test	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of defectives allowed 1/
Group I 2/				
Dielectric withstanding voltage	3.6	4.7.2	102	0
High voltage stabilization	3.11	4.7.7		
Group II 2/				
Visual and mechanical examination 3/	3.1, 3.4 to 3.5.2 incl; 3.23 to 3.25.1 incl	4.7.1	102 4/	0
Barometric pressure	3.7	4.7.3		
Insulation resistance	3.8	4.7.4		
Capacitance	3.9	4.7.5		
Dissipation factor	3.10	4.7.6		
Group III				1
Solderability	3.12	4.7.8	18	1
Vibration, high frequency	3.13	4.7.9		
Temperature coefficient and capacitance drift	3.14	4.7.10		
Temperature cycling and immersion	3.15	4.7.11		
Group IV				
Shock, specified pulse	3.16	4.7.12	18	1
Terminal strength	3.17	4.7.13		
Moisture resistance	3.18	4.7.14		
Group V				
Life (rated conditions)	3.19	4.7.15.1.1	50	1 6/
Life (accelerated conditions)	3.19	4.7.15.1.2	6	
Group VI				
Fungus 5/	3.20	4.7.16	5	0
Resistance to solvents	3.21	4.7.17	5	
Resistance to soldering heat	3.22	4.7.18		

1/ A sample unit having one or more defects shall be considered as a single defective.

2/ Nondestructive tests.

3/ Marking defects are based on visual examination only and shall be charged only for illegible, incomplete, or incorrect marking.

4/ Sample units shall be selected from those which have passed group I inspection.

5/ Certification of fungus resistance may be substituted for testing.

6/ For level "L": Maximum numbered failures allowed shall be based on the cumulative unit hours required for verification of 2 percent/1,000 hours failure rate level.

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4.5 Verification of qualification. Every 6 months, within 30 days of completion of group C inspection, the manufacturer shall compile a summary of the results of quality conformance inspections and extended FR test data, in the form of a Verification of Qualification Report, and forward it to the qualifying activity as the basis of continued qualification approval. In addition to the periodic submission of test data, the manufacturer shall immediately notify the qualifying activity whenever the data indicates that the manufacturer has failed to maintain his qualified FR level. Continued qualification approval will be based on evidence that, over the 6-month period, the following has been met:

(a) Verification by the qualifying activity that the manufacturer meets the requirements of MIL-STD-790.
(b) The manufacturer has not modified the design of the item.
(c) The specification requirements for the item have not been amended so as to affect the character of the item.
(d) Lot rejection for group A or B inspection does not exceed 10 percent or one lot, whichever is greater.
(e) The requirements of group C inspection are met.
(f) The records of all tests combined substantiate the "L" (2.0 percent), "M" (1.0 percent), "P" (0.1 percent) FR level has been maintained or that the manufacturer continued to meet the "R" (0.01 percent), "S" (0.001 percent), or "T" (0.0001 percent), FR level for which qualified, although the total component hours of testing does not, as yet, meet the requirements of 4.4.4(c).

If group C test requirements were not met and the manufacturer has taken corrective action satisfactory to the Government, the manufacturer shall submit a Verification of Qualification Report within 30 days after completion of the group C testing. In this case, the qualifying activity shall be notified within the time the original Verification of Qualification Report was due. All reports shall be certified by the responsible company official. The qualifying activity shall be contacted for report format.

4.5.1 *Records.* Maintenance of life test and FR level records shall be as specified in MIL-STD-690.

4.6 Quality conformance inspection.

4.6.1 *Inspection of product for delivery.* Inspection of product for delivery shall consist of group A inspection for FR levels M, P, R, S, and T or groups A and B inspections for FR level L.

4.6.1.1 *Inspection lot.* An inspection lot shall be as defined in MIL-STD-105, and shall consist of capacitors from the same style, production line or lines, and produced during a maximum period of one month. Each lot shall be kept separate from every other lot. The sample selected from the lot shall be representative of the case size in the lot, with the units being selected for test having the highest capacitance value in each case size. In addition, the following details shall apply:

(a) Styles CMR06, CMR07, and CMR08 may be combined.
(b) Styles CMR04 and CMR05 shall be kept separate from each other and from all other styles.
(c) All voltages within a style shall be kept separate.

4.6.1.2 *Group A inspection.* Group A inspection shall consist of the tests and examination specified in table VIII.

4.6.1.2.1 *Sampling plan.* Except for subgroup 1, statistical sampling and inspection shall be in accordance with MIL-STD-105 using general inspection levels. The tests in subgroup I shall be performed on each capacitor offered for inspection. The acceptable quality levels (AQL) for subgroups 2 and 3 shall be as specified in table VIII. The degree of sampling (normal or tightened) shall be determined independently of each subgroup.

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TABLE VIII. *Group A inspection.*

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)
Subgroup 1			
Dielectric withstanding voltage	3.6	4.7.2	Not applicable (100% insp.)
High voltage stabilization	3.11	4.7.7	
Subgroup 2			
Insulation resistance	3.8	4.7.4	0.15
Capacitance	3.9	4.7.5	
Dissipation factor	3.10	4.7.6	
Subgroup 3			
Visual and mechanical examination	3.1, 3.4 to 3.5.2 incl, 3.23 to 3.25.1 incl	4.7.1	1.0

4.6.1.2.1.1 *Manufacturer's production inspection.* If the manufacturer performs tests equal to or more stringent than those specified in subgroup 1, table VIII, as the final step of his production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria shall be complied with:

(a) Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than that specified for subgroup 1. Test conditions shall be equal to or more stringent than those specified for subgroup 1 tests.
(b) Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.
(c) The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.
(d) The lot rejection criteria is the same or more stringent than that specified herein.
(e) The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests which are a substitute for subgroup 1 tests. This data shall be provided as part of the evaluation required for MIL-STD-790. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests which are a substitute for subgroup 1 tests.
(f) Once approved, the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity.

4.6.1.2.2 *Rejected lots.* Lots rejected by the group A inspection shall be segregated from new lots and those lots that have passed inspection. Lots rejected beause of failure in subgroups 2 and 3 may be offered for acceptance only if the manufacturer inspects all sample units in the lot for those quality characteristics found defective in the sample and, after removing all defective units found, reinspects the lot using the tightened inspection procedure of MIL-STD-105. If, during the 100-percent inspection of subgroup 1, screening requires that over 8 percent of the capacitors be discarded, the lot shall be rejected.

4.6.1.3 *Group B inspection.*

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4.6.1.3.1 *For "L" FR level.* Group B inspection for "L" FR level shall consist of the test specified in table IX, and shall be performed on sample units which have been subjected to and have passed group A inspection. The lot conformance FR procedures are as specified in Procedure IV of MIL-STD-690 with the following details and exceptions:

(a) Lot conformance plan - 21 sample units, minimum.
(b) Duration of lot conformance FR test - 240 +48 -0 hours.
(c) Failure criteria - See 3.19. One failure allowed.
(d) Permissible combinations - See 4.6.1.1.
(e) Disposition of samples and inspection lot - Samples shall not be delivered on contract or order; however, a minimum of 10 units shall be randomly selected from the sample units and subjected to the extended life test as specified in 4.6.1.3.3. The inspection lot may be shipped upon successful completion of lot conformance FR test.

TABLE IX. *Group B inspection (for L FR level).*

Test	Requirement paragraph	Method paragraph
Life (240 hours accelerated)	3.19	4.7.15.1

4.6.1.3.2 *For M, P, R, S, and T FR levels.* A minimum of 10 sample units shall be selected at random from each inspection lot and shall be inspected as specified in 4.6.2.1.1.2: Units selected may be subjected to specified test before one month has elapsed.

4.6.1.3.3 *Extended life test (for "L" FR level).* Sample units, which have been subjected to 240-hour life test, shall be subjected to an additional 1,760 hours of life test as specified in 4.7.15.3.2.

4.6.1.3.4 *Failure in group B inspection for "L" FR level.* If an inspection lot is rejected as a result of failure to pass group B inspection for "L" FR level the lot shall not be resubmitted. Even though the lot has been rejected, those units which were predesignated for extended life testing shall remain or be placed on test for the full length of time.

4.6.2 *Periodic inspection.*

4.6.2.1 *Group C inspection.* Group C inspection shall be performed on sample units which have been subjected to and have passed group A inspection for FR levels M, P, R, S, and T and groups A and B inspections for FR level L and shall consist of the tests specified in table X, in the order shown. The maximum and minimum case size manufactured during a month's production shall be represented in the sample in at least the approximate ratio of production. Test data obtained shall be reviewed as part of the complete verification of qualification.

4.6.2.1.1 *Sampling plan.*

4.6.2.1.1.1 *For all FR levels.* Seventy-two sample units, representative of a month's production, shall be taken from production every month, divided into subgroups as listed in table X for subgroups 1 and 2, and subjected to the tests specified therein. Allowable failures shall be as specified in table X.

4.6.2.1.1.2 *For M, P, R, S, and T FR levels.* In addition to the tests specified in 4.6.2.1.1.1, a minimum of 10 sample units from each inspection lot (see 4.6.1.3.2) shall be subjected to subgroup 3 of table X. Allowable failures shall be as specified in MIL-STD-690, table IV, maintenance period A.

4.6.2.1.2 *Failures.* If the number of failures exceeds the number allowed in table X or MIL-STD-690, as applicable, the sample shall be considered to have failed.

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TABLE X. *Group C inspection.*

Test	Requirement paragraph	Method paragraph	Number of sample units	Allowable failures
All FR levels				
Subgroup 1				
Solderability	3.12	4.7.8	36	1
Vibration, high frequency	3.13	4.7.9		
Temperature coefficient and capacitance drift	3.14	4.7.10		
Temperature cycling and immersion	3.15	4.7.11		
Subgroup 2				
Barometric pressure	3.7	4.7.3	36	1
Shock, specified pulse	3.16	4.7.12		
Terminal strength	3.17	4.7.13		
Moisture resistance	3.18	4.7.14		
M, P, R, S, and T FR levels				
Subgroup 3				
Life (2,000 hours accelerated)	3.19	4.7.15.2	10 1/	1/

1/ See 4.6.2.1.1.2.

4.6.2.1.3 *Disposition of sample units.* Sample units which have been subjected to group C inspection shall not be delivered on the contract or order.

4.6.2.1.4 *Noncompliance.* If a sample fails to pass group C inspection, the manufacturer shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B (for "L" FR level) inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 *Inspection of preparation for delivery.* Sample packages and packs and the inspection of the preservation and packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of MIL-C-39028.

4.7 Methods of examination and test.

4.7.1 *Visual and mechanical examination.* Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4 to 3.5.2 inclusive, and 3.23 to 3.25.1 inclusive).

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4.7.2 *Dielectric withstanding voltage (see 3.6).* Capacitors shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Magnitude and nature of test voltage - 200 percent of rated dc voltage.
(b) Duration of application of test voltage - Not less than 1 second nor more than 5 seconds. The duration of the test shall begin when 95 percent of the test potential is reached.
(c) Points of application of test voltage - Between terminals.
(d) Limiting value of surge current - Shall not exceed 50 milliamperes (ma) during charging and discharging.
(e) Examination after test - Capacitors shall be examined for evidence of damage, arcing, and breakdown.

4.7.3 *Barometric pressure (see 3.7).* Capacitors shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - Not applicable.
(b) Test condition - D (100,000 ft).
(c) Tests during subjection to reduced pressure - A test potential of 100 percent of rated dc voltage for style CMR04, 500 volts, and 150 percent for all other styles and voltages shall be applied between the terminals for 60 +15, -0 seconds. Surge current shall not exceed 50 ma.
(d) Examination after test - Capacitors shall be examined for evidence of damage, arcing, and breakdown.

4.7.4 *Insulation resistance.*

4.7.4.1 *At room ambient temperature (see 3.8.1).* Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

(a) Test condition - A. (At the option of the manufacturer, a higher voltage may be used.)
(b) Points of measurement - From terminal to terminal. (Condensed moisture may be removed by a blast of air.)
(c) Electrification time - Not less than 10 seconds and not more than 2 minutes.

4.7.4.2 *At high ambient temperature (see 3.8.2).* Capacitors shall be subjected to the high ambient temperature, 125° +5°, -0° C or 150° +5°, -0° C, as applicable, for a period of time sufficient to reach thermal stability and shall be measured as specified in 4.7.4.1.

4.7.5 *Capacitance (see 3.9).* Capacitance shall be measured in accordance with method 305 of MIL-STD-202. The following details shall apply:

(a) Test frequency - 1 megahertz (MHz) ±1,000 hertz (Hz) when the nominal capacitance is 1,000 pF or less, and 1 kilohertz (kHz) ±100 Hz when the nominal capacitance is greater than 1,000 pF. At the option of the manufacturer, capacitance measurements may be made at any frequency from 1 kHz to 1 MHz and referred to measurements at 1 MHz and 1 kHz, as applicable.
(b) Limit of accuracy - Shall be ±0.2 percent of nominal capacitance value or ±0.2 pF, whichever is greater.

4.7.6 *Dissipation factor (see 3.10).* Dissipation factor shall be measured at a frequency of 1 MHz ±1,000 Hz when the nominal capacitance is 1,000 pF or less, and 1 kHz ±100 Hz when the nominal capacitance is greater than 1,000 pF. Measurement accuracy shall be within ±2 percent for dissipation factor and within ±5 Hz for frequency.

4.7.7 *High voltage stabilization (see 3.11).* Capacitors shall be subjected to 200 percent of rated dc voltage at a minimum temperature of 125° or 150° C, as applicable, for a minimum of 48 hours. During the test, the

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capacitors shall be adequately protected against voltage surges of 25 percent or more of test voltage and surge current shall not exceed 50 ma. Capacitors shall be examined for evidence of damage, arcing, and breakdown. Dielectric withstanding voltage and insulation resistance shall then be measured as specified in 4.7.2 and 4.7.4.1, respectively.

4.7.8 *Solderability (see 3.12).* Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

(a) Number of terminations of each part to be tested - Two.
(b) Depth of immersion in flux and solder- To a point at which exposed metal is solderable.

4.7.9 *Vibration, high frequency (see 3.13).* Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - Capacitors shall be rigidly mounted by the body to a vibration-test apparatus.
(b) Test condition - B.
(c) Measurements during vibration- During the last cycle in each direction, an electrical measurement shall be made to determine intermittent contacts of 0.5 ms or greater duration or open- or short-circuiting.
(d) Examination and measurements after test - Capacitors shall be examined for evidence of mechanical damage.

4.7.10 *Temperature coefficient and capacitance drift (see 3.14).*

4.7.10.1 *For qualification inspection.* Capacitance measurements shall be made in accordance with 4.7.5 (a frequency of 100 kHz ±10 percent may be used as an alternate) at the following temperatures in the order listed:

Step	Temperature °C	Step	Temperature °C
1	25 ±2	6	45 ±2
2	-55 +0, -2	7	65 ±2
3	-40 ±2	8	85 +2, -0
4	-10 ±2	9	125 +2, -0 (for temperature ranges "O" and "P")
5	25 ±2	10	150 +2, -0 (for temperature range "P")
		11	25 ±2

The measurement at each temperature shall be recorded when two successive readings taken at 5-minute intervals at that temperature indicate no change in capacitance. An accuracy of ±0.025 percent of nominal capacitance +0.05 pF shall be maintained for measurement of capacitance change.

4.7.10.1.1 *Temperature coefficient.* The temperature coefficient shall be computed as follows:

$$TC = \frac{(C_X - C_5)\,10^6}{(T_2 - T_1)\,C_5}$$

Where:
TC = Temperature coefficient in parts per million per degree centigrade.
C_X = Capacitance at test temperature in pF.
C_5 = Capacitance at step 5 temperature in pF.
T_1 = 25° C.
T_2 = Test temperature in degrees C.

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4.7.10.1.2 *Capacitance drift.* Capacitance drift shall be computed by dividing the greatest single difference between any two of the three values recorded at 25° C by the step 5 value recorded at 25° C.

4.7.10.2 *For quality conformance inspection.* Capacitance measurements shall be made as specified in 4.7.10.1, except that measurements shall be made only for steps 1, 2, 5, 9 or 10, as applicable, and 11.

4.7.10.3 *Continuous-curve temperature coefficient.* As an alternate to the measurements specified in 4.7.10.1, a continuous curve of capacitance versus temperature may be produced by subjecting the capacitors to a slowly varying temperature. The temperature shall be varied from 25° C to -55° C, to 125° or 150° C, as applicable, and to 25° C. A temperature-sensing device shall be embedded in a dummy capacitor in a manner to assure accurate internal readings in the capacitor under test. The temperature shall be varied slowly enough to produce a smooth, uniform curve with no loops at -55° C or 125° or 150° C, as applicable. Measurements shall be made at a frequency of 100 kHz ±10 kHz. Accuracy shall be as specified in 4.7.10.1.

4.7.11 *Temperature cycling and immersion (see 3.15).*

4.7.11.1 *Temperature cycling.* Capacitors shall be tested in accordance with method 102 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition - D, except that elevated temperature shall be 125° +3°, -0° or 150° +3°, -0° C, as applicable.
(b) Measurement before cycling - Capacitance.
(c) Measurement after cycling - Not applicable.

4.7.11.2 *Immersion.* The temperature cycling shall be followed by immersion cycling in accordance with method 104, test condition B, of MIL-STD-202. Capacitors shall be visually examined for evidence of deterioration, permanent damage to leads and case, and corrosion on the leads. Dielectric withstanding voltage, insulation resistance, capacitance, and dissipation factor shall then be measured as specified in 4.7.2, 4.7.4.1, 4.7.5, and 4.7.6, respectively.

4.7.12 *Shock, specified pulse (see 3.16).* Capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Mounting - Capacitors shall be rigidly mounted by the body.
(b) Test condition - I (100 G (pk), sawtooth).
(c) Measurements during shock - During the last shock in each direction, an electrical measurement shall be made to determine intermittent contacts of 0.5 ms or greater duration or open- or short-circuiting.
(d) Measurements after shock - Capacitors shall be examined for evidence of mechanical damage.

4.7.13 *Terminal strength (see 3.17).*

4.7.13.1 *Pull test.* All terminals shall be subjected for 5 ±1 seconds, to a pull of 5-pounds +2, -0 ounces in a direction perpendicular to the leads.

4.7.13.2 *Twist test.* Capacitors shall be tested in accordance with method 211, test condition D, of MIL-STD-202.

4.7.14 *Moisture resistance (see 3.18).* Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Number of cycles - Capacitors shall be subjected to twenty continuous temperature cycles.
(b) Mounting - Capacitors shall be mounted by the leads in a manner that will keep the capacitor body from touching the test fixture during test or measurement.

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(c) Conditioning prior to test - Capacitors shall be tested as specified in 4.7.11.1.
(d) Polarization - A dc potential of 100 volts shall be applied across the capacitor terminals of 50 percent of the units under test and no voltage applied to the other 50 percent.
(e) Vibration - The vibration requirement of steps 7a and 7b shall not apply.
(f) Final measurements - After completion of step 6 of the final cycle, capacitors shall be maintained at 25° ±5° C, and 50 ±5 percent relative humidity for a period of 4 to 24 hours. Insulation resistance, dielectric withstanding voltage, capacitance, and dissipation factor shall then be measured as specified in 4.7.4.1, 4.7.2, 4.7.5, and 4.7.6, respectively. Capacitors shall then be examined for evidence of deterioration, permanent damage to the leads or case, and corrosion on the leads.

4.7.15 *Life.* Capacitors shall be preconditioned for life test by subjecting them to a temperature of -55° +0°, -2° C for a minimum period of 48 hours. Capacitors shall then be subjected to the tests specified in 4.7.15.1 or 4.7.15.2, as applicable.

4.7.15.1 *For qualification inspection (2,000-hour) (see 3.19).*

4.7.15.1.1 *Rated conditions.* Capacitors shall be subjected to 100 percent of rated dc voltage at 125° ±3° C or 150° ±3° C, as applicable, for 2,000 +72 -0 hours. During the test, the surge current shall be limited to 50 milliamperes. At the conclusion of the test, capacitors shall be returned to the inspection conditions specified in 4.3.1, and shall be visually examined for evidence of mechanical damage; the dielectric withstanding voltage, insulation resistance, capacitance and dissipation factor shall then be measured as specified in 4.7.2, 4.7.4, 4.7.5 and 4.7.6, respectively (see 3.19).

4.7.15.1.2 *Accelerated conditions.* Capacitors shall be tested as specified in 4.7.15 and 4.7.15.1.1, except that the capacitors shall be subjected to 150 percent of rated dc voltage.

4.7.15.2 *For quality conformance inspection (see 3.19).*

4.7.15.2.1 *For FR level L.* Capacitors shall be tested as specified in 4.7.15 and 4.7.15.1.1, except that the capacitors shall be subjected to 150 percent of rated dc voltage for 240 +48, -0 hours.

4.7.15.2.2 *For FR levels M, P, R, S, and T.* Capacitors shall be tested as specified in 4.7.15.1.2.

4.7.15.3 *Extended life (see 3.19).*

4.7.15.3.1 *For qualification inspection.* Sample units which have been subjected to 2,000 hours of life test at rated conditions, as specified in 4.7.15.1.1, shall remain on test for an additional 48,000 +96, -0 hours. The measurements specified in 4.7.15.1.1 shall be made after every 2,000 hours for the first 10,000 hours and then after every 5,000 hours until a total of 50,000 hours has elapsed.

4.7.15.3.2 *For quality conformance inspection.* Sample units which have been subjected to 240 hours of life test, as specified in 4.7.15.2.1, and predesignated for an extended life test shall remain on test for an additional 1,760 +72, -0 hours. The measurements specified in 4.7.15.1.1 shall be made at the conclusion of 2,000 hours of life test.

4.7.16 *Fungus (see 3.20).* Capacitors shall be tested in accordance with method 508 of MIL-STD-810. Pretest and post-test measurements are not required.

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4.7.17 *Resistance to solvents (see 3.21).* Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) Portion of specimen to be brushed - That portion on which marking is present.
(b) Number of specimens to be tested - As specified in table VII.
(c) Permissible extent of damage - As specified in 3.21.

4.7.18 *Resistance to soldering heat (see 3.22).* Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

(a) Depth of immersion - Within .250-inch of the case.
(b) Test condition - B (260 ±5° C; 10 ±1 seconds).
(c) Cooling time prior to final measurement - 1 minute ±10 seconds.
(d) Measurements after test - Insulation resistance, capacitance and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively.

5. PREPARATION FOR DELIVERY

5.1 Capacitors shall be prepared for delivery in accordance with MIL-C-39028. In addition to any special marking required by the contract or purchase order (see 6.2), each unit package shall be marked with the source and date codes (see 3.23).

6. NOTES

6.1 Intended use. The capacitors covered by this specification are intended for use in equipment where a known order of reliability is required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the complete type designation (see 1.2.1 and 3.1).
(c) Levels of preservation and packaging and packing, and applicable marking (see section 5).
(d) Marking requirement for failure rate level (see 3.24).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is Army Electronics Command; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45401.

6.3.1 *Failure rate level qualification.* Where the invitation for bids specifies a failure rate level below the "P" level, awards will be made only for those products that have been qualified prior to the bid-opening date by the qualifying activity as meeting the failure rate level specified in the invitation for bids.

6.4 Selection and use information. Equipment designers should refer to MIL-STD-198, "Capacitors, Selection and Use of" for a selection of standard capacitor types and values for new equipment design. Application and use information concerning these capacitors are also provided in MIL-STD-198.

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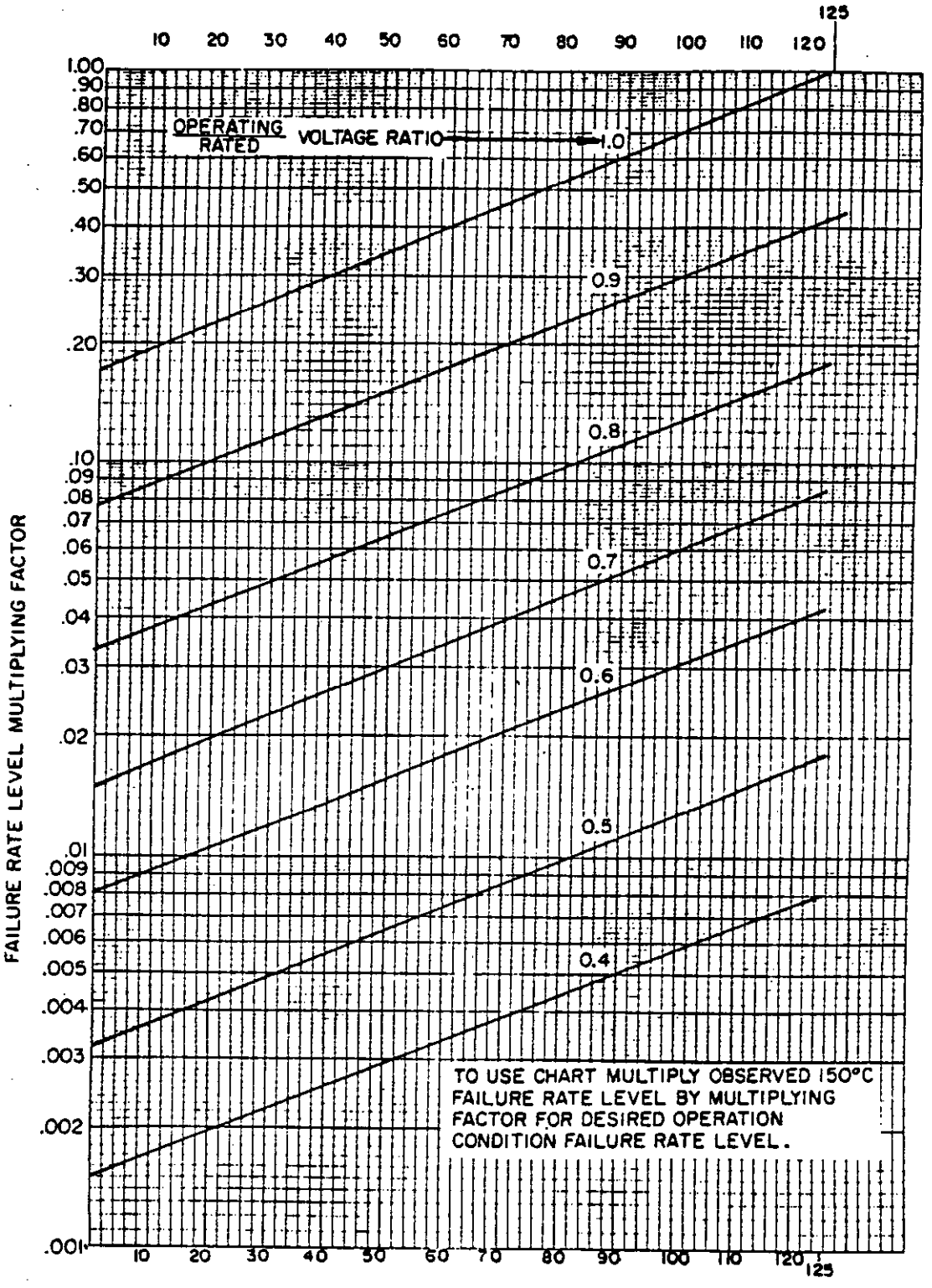


FIGURE 3. Failure rate curves; 0 - to 125°C.

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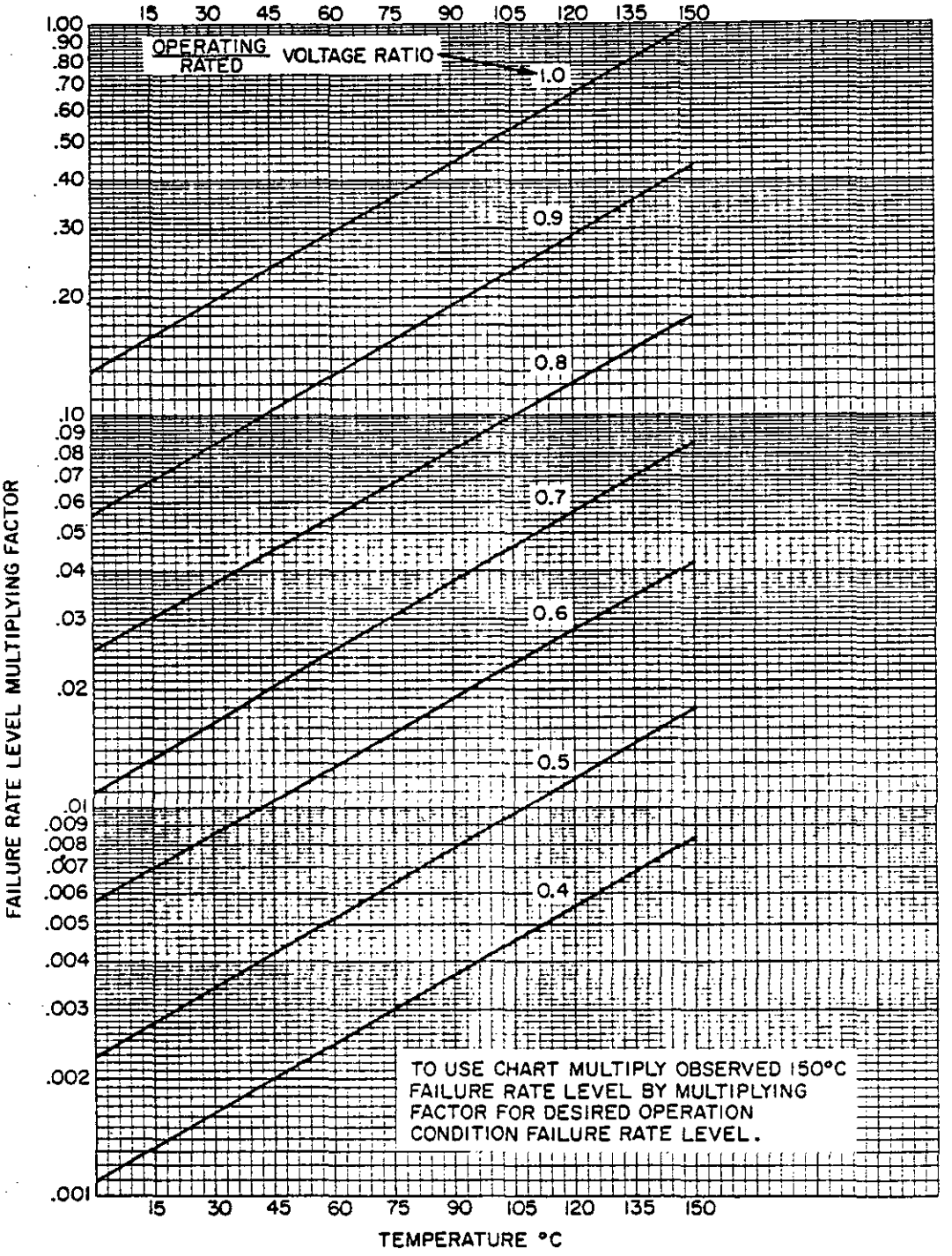


FIGURE 4. Failure rate curves; 0 - to 150° C.

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6.5 Failure-rate level determination. The curves presented on figures 3 and 4 are the best engineering approximation of the reliability characteristics (random failures) for these capacitors when employed repeatedly, within their specification ratings, in complex electronic equipment. These reliability characteristics are based on ground-level severity experience. Failures are considered to be opens, shorts, or radical departures from initial characteristics. The failures are considered to be occurring in an unpredictable manner and in too short a period of time to permit detection through normal preventive maintenance. The curves shown on this figure are based on "catastrophic failures" and will differ from the failure rates established in the specification, since the established failure rates are based on "parameter failures" over long term life tests at rated conditions. Figures 3 and 4 have been extracted from MIL-HDBK-217, "Reliability Stress and Failure Rate Data for Electronic Equipment." The curves have been modified from their original versions in that the ordinate has been normalized in order to provide multiplier factors in place of discrete failure rate levels and in order that the multiplying factor for a failure rate at rated conditions is unity. As indicated, these curves are the best estimates based on "catastrophic failures"; however, they can provide an estimate of the relative effect of operating under conditions other than rated.

6.6 **Changes from previous issue.** Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 This appendix details the procedure for submission of samples, with related data, for initial qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined herein.

20. SUBMISSION

20.1 Sample.

20.1.1 *Single-type submission.* A sample consisting of 102 sample units of each type for which qualification is sought shall be submitted (see 30.1).

20.1.2 *Single-style submission.* A maximum of two types, 51 of each type, may be represented in a single style submission. Both types shall be of the same capacitance tolerance and characteristic. The highest and lowest capacitance value within a style or characteristic shall be equally represented. However, where the lowest capacitance value listed for a particular style or characteristic is below 47 pF, specimens of the 47 pF value, or preferably lower, may be submitted.

20.2 Test data. When examinations and tests are to be performed at a Government laboratory, prior to submission, all sample units shall be subjected to all of the examinations and tests indicated as nondestructive in table VII. Each submission shall be accompanied by the test record obtained from these examinations and tests. The performance of the destructive tests by the manufacturer on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

20.3 Certification of material. When submitting samples for qualification, the manufacturer shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

20.4 Description of items. The manufacturer shall submit a detailed description of the materials and general constructional features of the complete range of capacitors for which qualification is sought; for example, case material, type of electrodes and lead metal, dielectric thickness, margin, etc.

30. EXTENT OF QUALIFICATION

30.1 Single-type submission. Qualification will be restricted to the style, capacitance value, characteristic, and capacitance tolerance equal to or broader than the characteristic and capacitance tolerance submitted.

30.2 Single-style submission. Qualification of all types in the submission automatically carries with it qualification of all types within the style and capacitance range submitted, except that approval of the lowest capacitance value in the "F" characteristic will be the basis for approval of all lesser capacitance values and broader characteristics within a style. Qualification of characteristics and capacitance tolerances shall be restricted to those equal to or broader than that submitted. This extension of qualification is contingent on the use of the same materials, design, and construction for all types qualified. Wherever material, design, or constructional differences exist, submission shall be made in accordance with 20.1.1.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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